

ISyE6669 Homework 7

Fall 2025

1 Week 7

1. Geometry of LP

- (a) Let P be a triangle in a plane with three extreme points $A = [1, 1]^\top$, $B = [2, 4]^\top$, $C = [4, 2]^\top$. Let $x = [2, 2]^\top$ be a point in the triangle P . Express x as a convex combination of the three extreme points A, B, C . You should write $x = \lambda_1 A + \lambda_2 B + \lambda_3 C$, where $\lambda_1 + \lambda_2 + \lambda_3 = 1$ and all nonnegative. You need to give the numerical values of $\lambda_1, \lambda_2, \lambda_3$.

Solution: By plugging given values into following equations:

$$x_1 = \lambda_1 x_{1,A} + \lambda_2 x_{1,B} + \lambda_3 x_{1,C}$$

$$x_2 = \lambda_1 x_{2,A} + \lambda_2 x_{2,B} + \lambda_3 x_{2,C}$$

$$1 = \lambda_1 + \lambda_2 + \lambda_3$$

$$\lambda_i \geq 0, i = \{1, 2, 3\}$$

we obtain following system of equations:

$$2 = \lambda_1 + 2\lambda_2 + 4\lambda_3$$

$$2 = \lambda_1 + 4\lambda_2 + 2\lambda_3$$

$$1 = \lambda_1 + \lambda_2 + \lambda_3$$

$$\lambda_i \geq 0, i = \{1, 2, 3\}$$

By subtracting third equation from the first and the second we get:

$$1 = \lambda_2 + 3\lambda_3$$

$$1 = 3\lambda_2 + \lambda_3$$

$$1 = \lambda_1 + \lambda_2 + \lambda_3$$

$$\lambda_i \geq 0, i = \{1, 2, 3\}$$

By multiplying the first equation by 3 and subtracting from the third, we get:

$$\begin{aligned}
1 &= \lambda_2 + 3\lambda_3 \\
-2 &= -8\lambda_3 \\
1 &= \lambda_1 + \lambda_2 + \lambda_3 \\
\lambda_i &\geq 0, i = \{1, 2, 3\}
\end{aligned}$$

Now, from the second equation we get

$$\lambda_3 = \frac{1}{4}.$$

From the first equation we get:

$$\lambda_2 = 1 - 3\lambda_3 = \frac{1}{4}$$

And finally, from the third equation we get:

$$\lambda_1 = 1 - \lambda_2 - \lambda_3 = \frac{1}{2}$$

Thus $x = \frac{1}{2}A + \frac{1}{4}B + \frac{1}{4}C$.

(b) Consider the following polyhedron P defined as

$$P = \{(x, y) \in \mathbb{R}^2 : x + 2y \geq 3, x - y \leq 1, y \geq 1, x \leq -1\}.$$

Find all the extreme points and extreme rays of P .

Solution: We can graph polyhedron P

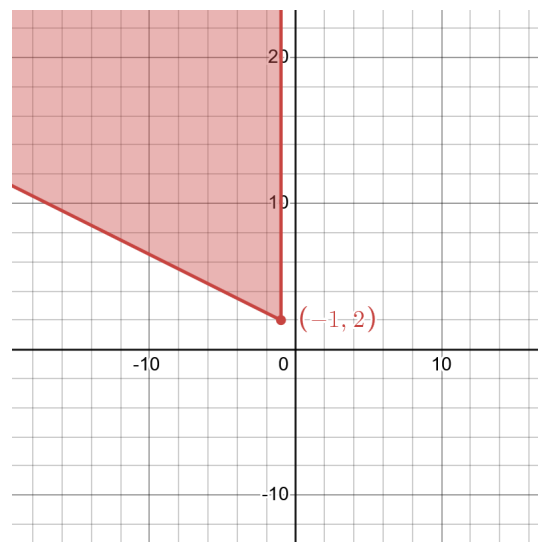


Figure 1: Polyhedron P .

and from the graph we see that it has one extreme point $(-1, 2)$, and two extreme rays $(-2, 1)$ and $(0, 1)$.

We can also find extreme points and extreme rays algebraically: We generate the extreme points by finding basic solutions of $n = 2$ linearly independent active constraints and verifying their feasibility:

i. By solving system:

$$\begin{cases} 3 = x + 2y \\ 1 = x - y \end{cases}$$

we get $(5/3, 2/3)$. This point is not feasible because it does not satisfy $x \leq -1$.

ii. By solving system:

$$\begin{cases} 3 = x + 2y \\ 1 = y \end{cases}$$

we get $(1, 1)$. This point is not feasible because it does not satisfy $x \leq -1$.

iii. By solving system:

$$\begin{cases} 3 = x + 2y \\ -1 = x \end{cases}$$

we get $(-1, 2)$. This point is feasible, but it is the same extreme point we already found.

iv. By solving system:

$$\begin{cases} 1 = x - y \\ 1 = y \end{cases}$$

we get $(2, 1)$. This point is not feasible because it does not satisfy $x \leq -1$.

v. By solving system:

$$\begin{cases} 1 = x - y \\ -1 = x \end{cases}$$

we get $(-1, -2)$. This point is not feasible because it does not satisfy $y \geq 1$.

vi. By solving system:

$$\begin{cases} 1 = y \\ -1 = x \end{cases}$$

we get $(-1, 1)$. This point is not feasible because it does not satisfy $x + 2y \geq 3$.

To find extreme rays of P we need to find extreme directions of recession cone $\{(x, y) \in \mathbb{R}^2 : x + 2y \geq 0, x - y \leq 0, y \geq 0, x \leq 0\}$. We generate the extreme directions by finding basic solutions of $n = 1$ linearly independent active constraints and verifying their feasibility:

i. By solving

$$x + 2y = 0$$

we get solutions $(-2\alpha, \alpha) = \alpha(-2, 1)$, $\alpha \in \mathbb{R}$. From the constraints we see that these solutions are feasible for all $\alpha \geq 0$. Hence $(-2, 1)$ is an extreme ray/direction.

ii. By solving

$$x - y = 0$$

we get solutions $(\alpha, \alpha) = \alpha(1, 1)$, $\alpha \in \mathbb{R}$. From the constraints we see that these solutions are feasible only for $\alpha = 0$. Hence $(1, 1)$ is not an extreme ray/direction.

iii. By solving

$$y = 0$$

we get solutions $(\alpha, 0) = \alpha(1, 0)$, $\alpha \in \mathbb{R}$. From the constraints we see that these solutions are feasible only for $\alpha = 0$. Hence $(1, 0)$ is not an extreme ray/direction.

iv. By solving

$$x = 0$$

we get solutions $(0, \alpha) = \alpha(0, 1)$, $\alpha \in \mathbb{R}$. From the constraints we see that these solutions are feasible for all $\alpha \geq 0$. Hence $(0, 1)$ is an extreme ray/direction.

- (c) Remember that a polytope is a bounded polyhedron. Give an example of a polytope in dimension n defined by exactly $2n$ inequalities, such that removing any inequality describing the polytope makes the resulting polyhedron unbounded. Hint: First think of an example in $\mathbb{R}^2$ and then generalize it.

By the way, an interesting fact is that examples of the above kind do not occur if the number of inequalities is more than $2n$. For example if a polytope is in dimension n and is defined by $2n + 1$ inequalities, then there always exists one inequality such that removing this inequality leaves the resulting (possibly larger) polyhedron bounded. I encourage you to experiment in $n = 2$ dimension to see this fact!

Solution: Example of a polytope in $\mathbb{R}^n$ defined using $2n$ constraints, such that removing any one constraint will result in unbounded polyhedron is a box, i.e. polytope defined using simple bounding constraints on every variable:

$$P = \{x = (x_1, \dots, x_n) \in \mathbb{R}^n : m_i \leq x_i \leq M_i, \forall i = 1, \dots, n\}$$

where $m = (m_1, \dots, m_n)$ and $M = (M_1, \dots, M_n)$ are constants.

2. Reformulation as linear program

- (a) Consider the following linear program:

$$\begin{aligned} \max \quad & x + 2y \\ \text{s.t.} \quad & x + y \leq 4, \\ & 2x - y \geq -2, \\ & 2x + y \leq 6, \\ & x \leq 0. \end{aligned}$$

Convert the above LP to a standard form LP.

Solution: We can reformulate above problem as standard form LP converting it to minimization and by introducing new variables x' , y^+ , y^- , s_1, s_2 and s_3 such that $x = -x'$, $y = y^+ - y^-$ and $s_1, s_2, s_3 \geq 0$. Now problem can be formulated as

$$\begin{aligned} \min \quad & x' - 2y^+ + 2y^- \\ \text{s.t.} \quad & -x' + y^+ - y^- + s_1 = 4, \\ & -2x' - y^+ + y^- - s_2 = -2, \\ & -2x' + y^+ - y^- + s_3 = 6, \\ & x', y^+, y^-, s_1, s_2, s_3 \geq 0. \end{aligned}$$

- (b) Alexis Cornby makes her living buying and selling corn. On January 1, she has 50 tons of corn and \$1000. On the first day of each month Alexis can buy corn at the following prices per ton: January \$300; February, \$350; March, \$400; April, \$500. On the last day of each month, Alexis can sell corn at following prices per ton: January, \$250; February, \$400; March, \$350; April, \$550. Alexis stores her corn in a warehouse that can hold at most 100 tons of corn. She must be able to pay cash for all corn at the time of purchase. Write a linear programming model that can be solved to determine how Alexis can maximize her cash on hand at the end of April. To get an idea of constraints and objective, let's consider a timeline of events ending with end of April:

- Start:
 - Action: None
 - Corn: 50 ton
 - Money: \$1,000
- January 1st:
 - Action: Buy x_J tons of corn
 - Corn: $50 + x_J$
 - Money: $\$1,000 - \$300x_J$
- : January 31st
 - Action: Sell y_J tons of corn
 - Corn: $50 + x_J - y_J$
 - Money: $\$1,000 - \$300x_J + \$250y_J$
- February 1st:
 - Action: Buy x_F tons of corn
 - Corn: $50 + x_J - y_J + x_F$
 - Money: $\$1,000 - \$300x_J + \$250y_J - \$350x_F$
- : February 28th/29th:
 - Action: Sell y_F tons of corn
 - Corn: $50 + x_J - y_J + x_F - y_F$
 - Money: $\$1,000 - \$300x_J + \$250y_J - \$350x_F + \$400y_F$
- March 1st:
 - Action: Buy x_M tons of corn

- Corn: $50 + x_J - y_J + x_F - y_F + x_M$
- Money: $\$1,000 - \$300x_J + \$250y_J - \$350x_F + \$400y_F - \$400x_M$
- March 31st:
 - Action: Sell y_M tons of corn
 - Corn: $50 + x_J - y_J + x_F - y_F + x_M - y_M$
 - Money: $\$1,000 - \$300x_J + \$250y_J - \$350x_F + \$400y_F - \$400x_M + \$350y_M$
- April 1st:
 - Action: Buy x_A tons of corn
 - Corn: $50 + x_J - y_J + x_F - y_F + x_M - y_M + x_A$
 - Money: $\$1,000 - \$300x_J + \$250y_J - \$350x_F + \$400y_F - \$400x_M + \$350y_M - \$500x_A$
- April 30th:
 - Action: Sell y_A tons of corn
 - Corn: $50 + x_J - y_J + x_F - y_F + x_M - y_M + x_A - y_A$
 - Money: $\$1,000 - \$300x_J + \$250y_J - \$350x_F + \$400y_F - \$400x_M + \$350y_M - \$500x_A + \$550y_A$

Now let's consider what are limitations Alexis has: she obviously can't buy or sell negative amounts of corn, every time she buys corn she has to ensure she has funds to pay for the corn and has enough storage for the corn she bought, and every time she sells the corn she has to ensure she does not sell more corn than what she has in her storage. Let's write down those constraints

Constraints:

- Sufficient funds to buy:
 - January 1st: $\$1000 - \$300x_J \geq 0$
 - February 1st: $\$1,000 - \$300x_J + \$250y_J - \$350x_F \geq 0$
 - March 1st: $\$1,000 - \$300x_J + \$250y_J - \$350x_F + \$400y_F - \$400x_M \geq 0$
 - April 1st: $\$1,000 - \$300x_J + \$250y_J - \$350x_F + \$400y_F - \$400x_M + \$350y_M - \$500x_A \geq 0$
- Sufficient storage for bought corn:
 - January 1st: $50 + x_J \leq 100$
 - February 1st: $50 + x_J - y_J + x_F \leq 100$
 - March 1st: $50 + x_J - y_J + x_F - y_F + x_M \leq 100$
 - April 1st: $50 + x_J - y_J + x_F - y_F + x_M - y_M + x_A \leq 100$
- No corn overdraft:
 - January 31st: $50 + x_J - y_J \geq 0$
 - February 28th/29th: $50 + x_J - y_J + x_F - y_F \geq 0$
 - March 31st: $50 + x_J - y_J + x_F - y_F + x_M - y_M \geq 0$
 - April 30th: $50 + x_J - y_J + x_F - y_F + x_M - y_M + x_A - y_A \geq 0$
- Non-negativity:
 - $x_i, y_i \geq 0, i \in \{J, F, M, A\}$

Alexis' goal is to maximize amount of cash she has on hand at the end of April so:

Objective: Maximize

$$1,000 - 300x_J + 250y_J - 350x_F + 400y_F - 400x_M + 350y_M - 500x_A + 550y_A$$

Finally we can put all this together to obtain following optimization problem:

$$\begin{aligned} \max \quad & 1,000 - 300x_J + 250y_J - 350x_F + 400y_F - 400x_M + 350y_M - 500x_A + 550y_A \\ \text{s.t.} \quad & 1000 - 300x_J \geq 0 \\ & 1,000 - 300x_J + 250y_J - 350x_F \geq 0 \\ & 1,000 - 300x_J + 250y_J - 350x_F + 400y_F - 400x_M \geq 0 \\ & 1,000 - 300x_J + 250y_J - 350x_F + 400y_F - 400x_M + 350y_M - 500x_A \geq 0 \\ & 50 + x_J \leq 100 \\ & 50 + x_J - y_J + x_F \leq 100 \\ & 50 + x_J - y_J + x_F - y_F + x_M \leq 100 \\ & 50 + x_J - y_J + x_F - y_F + x_M - y_M + x_A \leq 100 \\ & 50 + x_J - y_J \geq 0 \\ & 50 + x_J - y_J + x_F - y_F \geq 0 \\ & 50 + x_J - y_J + x_F - y_F + x_M - y_M \geq 0 \\ & 50 + x_J - y_J + x_F - y_F + x_M - y_M + x_A - y_A \geq 0 \\ & x_i \geq 0, \quad i \in \{J, F, M, A\} \\ & y_i \geq 0, \quad i \in \{J, F, M, A\} \end{aligned}$$

Solving problem was optional, but if you want to check your formulation, correct solution is following:

Cash on hand at the end of April: \$29,333.33

January:

- Buy: 3.33 tons
- Sell: 0 tons

February:

- Buy: 0 tons
- Sell: 15.45 tons

March:

- Buy: 15.45 tons
- Sell: 0 tons

April:

- Buy: 0 tons
- Sell: 53.33 tons

3. Basic Solution and Basic Feasible Solution

(a) Consider the following linear program:

$$\begin{aligned}
 \min \quad & -x_1 - 2x_2 \\
 \text{s.t.} \quad & x_1 - x_2 \leq 1 \\
 & 2x_1 - x_2 \geq -1 \\
 & 2x_1 + x_2 \leq 5 \\
 & x_1 \geq 0, \quad x_2 \geq 0.
 \end{aligned}$$

Graph the constraints of this linear program, and indicate the feasible region.

Solution: See Figure 2.

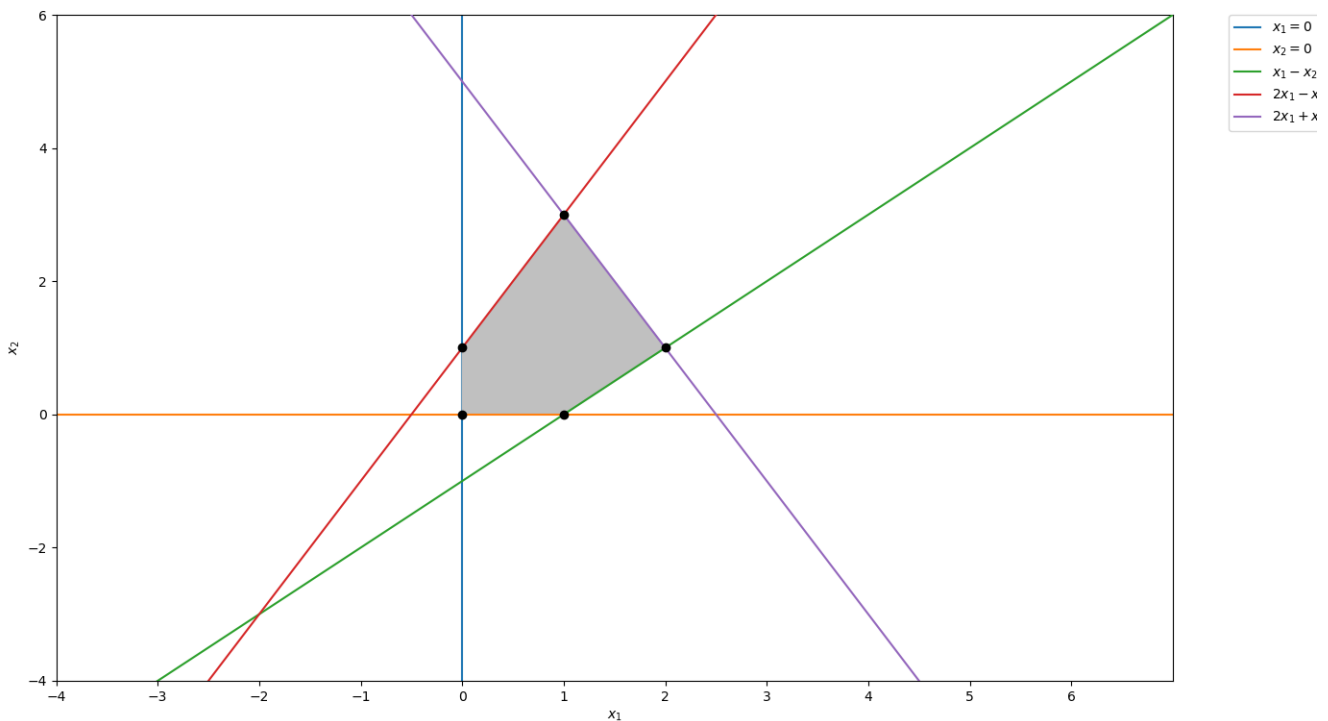


Figure 2: Feasible region

(b) Transform it into a standard form LP.

Denote $\mathbf{x} = [x_1, x_2, x_3, x_4, x_5]^\top$ as the vector of variables, and use the standard form notation:

$$\begin{aligned}
 \min \quad & \mathbf{c}^\top \mathbf{x} \\
 \text{s.t.} \quad & \mathbf{Ax} = \mathbf{b} \\
 & \mathbf{x} \geq \mathbf{0},
 \end{aligned}$$

specify $\mathbf{c}, \mathbf{A}, \mathbf{b}$ for the above problem.

Solution: We can rewrite the problem as:

$$\begin{aligned} \min \quad & -x_1 - 2x_2 \\ \text{s.t.} \quad & x_1 - x_2 + x_3 = 1 \\ & 2x_1 + x_2 - x_4 = -1 \\ & 2x_1 + x_2 + x_5 = 5 \\ & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, x_5 \geq 0. \end{aligned}$$

In this case, the $\mathbf{c}, \mathbf{A}$ and $\mathbf{b}$ are

$$\mathbf{c} = \begin{bmatrix} -1 \\ -2 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 1 & -1 & 1 & 0 & 0 \\ 2 & -1 & 0 & -1 & 0 \\ 2 & 1 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ -1 \\ 5 \end{bmatrix}.$$

- (c) Use the procedure discussed in lecture to find *all* basic solutions. For each basic solution, specify the basis matrix, the basic variables, the non-basic variables, and the associated cost coefficients for each part. For example, for the following basis matrix

$$\mathbf{B} = [\mathbf{A}_3, \mathbf{A}_4, \mathbf{A}_5] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

where $\mathbf{A}_i$ is the i -th column of $\mathbf{A}$, the corresponding basic variables, non-basic variables, and cost coefficients are

$$\mathbf{x}_B = \begin{bmatrix} x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} -1 \\ -2 \end{bmatrix}.$$

Specify all basic solutions in this way. [Hint: The nonbasic variables are always zero. So the key is to compute the basic variables. It will require inverting the basis matrix $\mathbf{B}$. You can use Python to compute the inverse of a matrix. Scipy has very fast linear algebra packages. You only need to invoke “from scipy import linalg” and use “linalg.inv(A)” to invert a matrix A .]

Solution: Below are all basic solutions for this problem:

$$\begin{aligned} (b1) \quad \mathbf{B}_2 &= [\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3] = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 2 & 1 & 0 \end{bmatrix}, \\ \mathbf{x}_B &= \mathbf{B}_2^{-1} \mathbf{b} = \begin{bmatrix} x_1 \\ x_2 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} x_3 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \end{aligned}$$

$$(b2) \quad \mathbf{B}_2 = [\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_4] = \begin{bmatrix} 1 & -1 & 0 \\ 2 & -1 & -1 \\ 2 & 1 & 0 \end{bmatrix},$$

$$\mathbf{x}_B = \mathbf{B}_2^{-1} \mathbf{b} = \begin{bmatrix} x_1 \\ x_2 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

$$(b3) \quad \mathbf{B}_3 = [\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_5] = \begin{bmatrix} 1 & -1 & 0 \\ 2 & -1 & 0 \\ 2 & 1 & 1 \end{bmatrix},$$

$$\mathbf{x}_B = \begin{bmatrix} x_1 \\ x_2 \\ x_5 \end{bmatrix} = \begin{bmatrix} -2 \\ -3 \\ 12 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

$$(b4) \quad \mathbf{B}_4 = [\mathbf{A}_1, \mathbf{A}_3, \mathbf{A}_4] = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 0 & -1 \\ 2 & 0 & 0 \end{bmatrix},$$

$$\mathbf{x}_B = \begin{bmatrix} x_1 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2.5 \\ -1.5 \\ 6 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} x_2 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} -2 \\ 0 \end{bmatrix}.$$

$$(b5) \quad \mathbf{B}_5 = [\mathbf{A}_1, \mathbf{A}_3, \mathbf{A}_5] = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 0 & 0 \\ 2 & 0 & 1 \end{bmatrix},$$

$$\mathbf{x}_B = \begin{bmatrix} x_1 \\ x_3 \\ x_5 \end{bmatrix} = \begin{bmatrix} -0.5 \\ 1.5 \\ 6 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} x_2 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} -2 \\ 0 \end{bmatrix}.$$

$$(b6) \quad \mathbf{B}_6 = [\mathbf{A}_1, \mathbf{A}_4, \mathbf{A}_5] = \begin{bmatrix} 1 & 0 & 0 \\ 2 & -1 & 0 \\ 2 & 0 & 1 \end{bmatrix},$$

$$\mathbf{x}_B = \begin{bmatrix} x_1 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} -2 \\ 0 \end{bmatrix}.$$

$$(b7) \quad \mathbf{B}_7 = [\mathbf{A}_2, \mathbf{A}_3, \mathbf{A}_4] = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix},$$

$$\mathbf{x}_B = \begin{bmatrix} x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ -4 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} x_1 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} -2 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

$$(b8) \quad \mathbf{B}_8 = [\mathbf{A}_2, \mathbf{A}_3, \mathbf{A}_5] = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix},$$

$$\mathbf{x}_B = \begin{bmatrix} x_2 \\ x_3 \\ x_5 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} x_1 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} -2 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

$$(b9) \quad \mathbf{B}_9 = [\mathbf{A}_2, \mathbf{A}_4, \mathbf{A}_5] = \begin{bmatrix} -1 & 0 & 0 \\ -1 & -1 & 0 \\ 1 & 0 & 1 \end{bmatrix},$$

$$\mathbf{x}_B = \begin{bmatrix} x_2 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} -2 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

$$(b10) \quad \mathbf{B}_{10} = [\mathbf{A}_3, \mathbf{A}_4, \mathbf{A}_5] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$\mathbf{x}_B = \begin{bmatrix} x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} -1 \\ -2 \end{bmatrix}.$$

- (d) Among all the basic solutions you found, which basic solutions are feasible, thus are basic feasible solutions? Which basic solutions are infeasible? Locate each basic solution on the graph you drew in part (a). [Hint: You only need to look at the (x_1, x_2) part of each basic solution.] What is the optimal solution?

Solution: Among solutions we found, solutions (b1), (b2), (b6), (b8), and (b10) are feasible thus are basic feasible solutions. They are represented as green dots in the Figure 3. Solutions (b3), (b4), (b5), (b7), and (b9) are infeasible. They are represented using red dots in the Figure 3. By calculating the objective value at each of the basic feasible solution we get that optimal solution is solution (b1), with an optimal value of -7 .

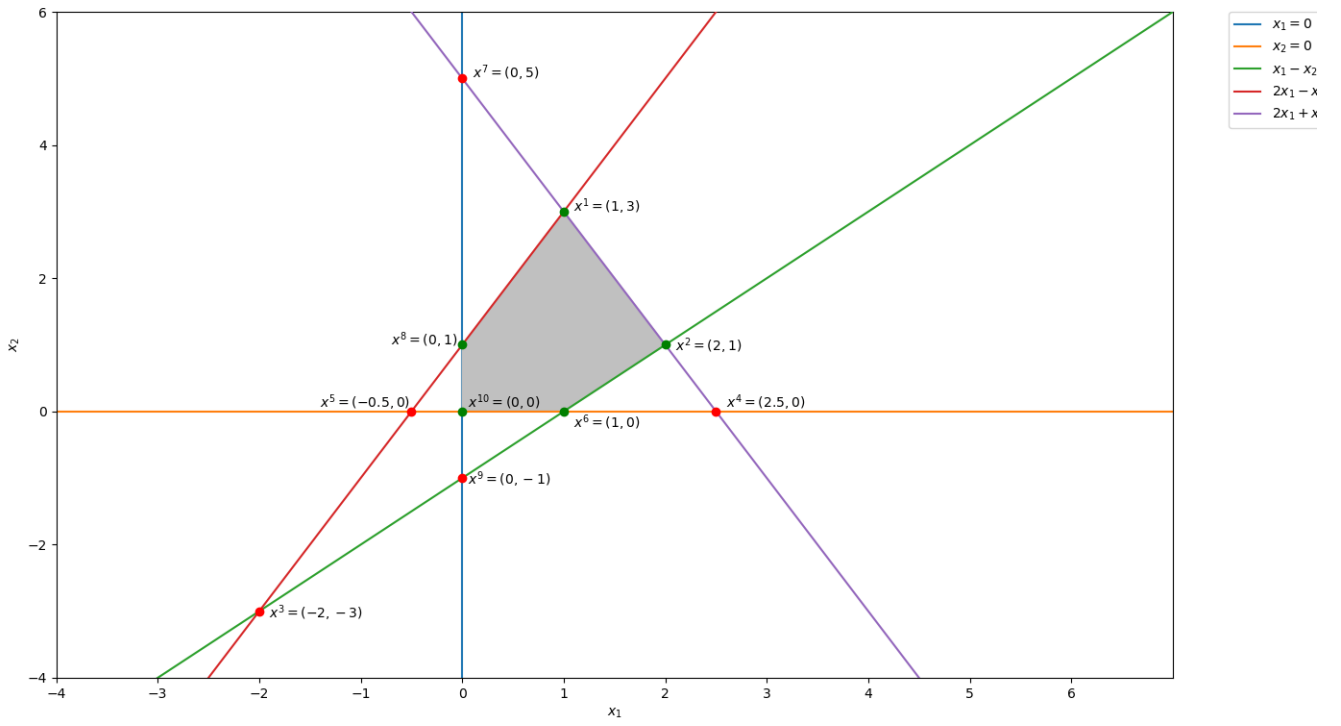


Figure 3: Basic solutions